

Robust Counting of Soft Fruit through Occlusions with Re-identification

Raymond Kirk¹, Michael Mangan², and Grzegorz Cielniak¹

¹ Lincoln Centre for Autonomous Systems, University of Lincoln, UK
`{rkirk,gcielniak}@lincoln.ac.uk`

² Dept of Computer Science, University of Sheffield, UK
`m.mangan@sheffield.ac.uk`

Abstract. Fruit counting and tracking is a crucial component of fruit harvesting and yield forecasting applications within horticulture. We present a novel multi-object, multi-class fruit tracking system to count fruit from image sequences. We first train a recurrent neural network (RNN) comprised of a feature extractor stem and two heads for re-identification and maturity classification. We apply the network to detected fruits in image sequences and utilise the output of both network heads to maintain track consistency and reduce intra-class false positives between maturity stages. The counting-by-tracking system is evaluated by comparing with a popular detect-to-track architecture and against manually labelled tracks (counts). Our proposed system achieves a mean average percentage error (MAPE) of 3% ($L1$ loss=7) improving on the baseline multi-object tracking approach which obtained an MAPE of 21% ($L1$ loss=41). Validating this approach for use in horticulture.

Keywords: Computer Vision · Multi-Object Tracking · Mobile Robotics

1 Introduction

Fruit counting is a critical process in effective management of a fruit crop. It informs decisions on harvesting, labour management and yield estimates. Labour constitutes for 65% of the total fruit harvesting cost and yield estimates typically have high uncertainty, motivating the need for accurate counting systems. With the advent of mobile agricultural robots and the success of convolutional neural network (CNN) based detectors, traditionally laborious tasks such as flower counting (a strong indicator of future yield) can now be automated.

Insight into the capabilities of a vision-based tracking system on a mobile robot is yet to be evaluated for fruit counting. Multi-object trackers in the detect-to-track paradigm have shown promise applied to people tracking, and some of these approaches have been successfully applied to fruit counting [1, 13, 18].

Tracking fruit, for the purpose of counting, faces many challenges due to the nature and complexity of farm environments. A tracking algorithm must be able to disambiguate near identical instances of fruit, handle changing appearances, manage varying factors such as illumination or altering-viewpoint, and



Fig. 1. Mobile robot tracking platform in the *Katrina-1* Strawberry row (left). Fruit counting-by-tracking (Ripe, Unripe and Flower) visualisation (right). The circles show the tracked fruit identities over time and tracks generated from our proposed method. Strawberry maturity classes are omitted for visualisation purposes, individual instances are in the format *TrackID_ClassID* where 1, 2, 3 are ripe, flower and unripe respectively.

re-identify after disappearances due to other issues such as occlusion. Examples are depicted in Fig. 1. Recently proposed solutions [4, 8] leverage deep learning to accurately detect fruit in varying conditions, with newer models subsequently adding detect-to-track based approaches as the counting method [11, 12]. These approaches generally deal with a single class for each fruit: we investigate the utilisation of fruit maturity stage and a mobile robot to enable more effective tracking.

A comparison of SOTA trackers [9] in the MOT challenge attribute the recent rise in tracker performance to the inclusion of stronger affinity and appearance models in tracking architectures. Enabling tracks to be maintained over more complex sequences. We explore this insight in this paper extending an appearance model-based tracking framework for improved counting of objects in horticultural environments.

The main contributions of our paper are: (1) a novel first re-identification and label probability based tracking framework applied on mobile robots for the purpose of counting fruits (2) extension of a popular re-identification tracking framework to embed contextual, shape and class information into association cost (3) four sequences of hand labelled Strawberry data for tracking in complex environments shared for bench-marking with the community and (4) validation of the counting accuracy for the purpose of yield estimation. A video of the tracking system and the code to run, train and reproduce the experiments can be found at the [fruit_tracking](#) repository.

2 Related Work

Recent object tracking frameworks follow the detect-to-track paradigm [2, 3, 15, 19] following the success of deep object detectors. Regional-based convolutional

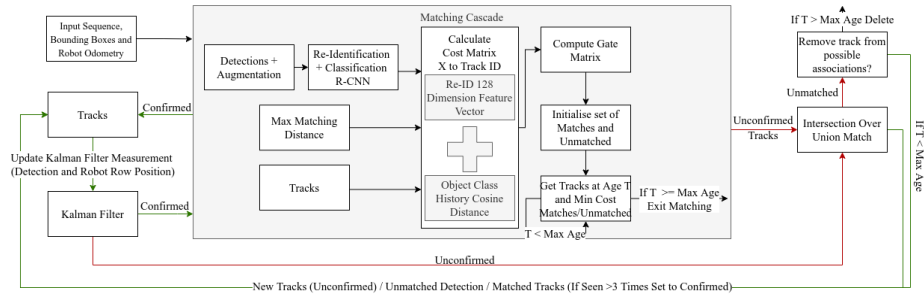


Fig. 2. The proposed fruit counting pipeline. The pipeline consists of three main stages, starting with the input of detections in the format of bounding boxes and the corresponding images, generation of re-identification feature vectors and class descriptors and then finally a matching cascade prioritising newer tracks with IoU matching.

neural networks (R-CNN) were introduced in 2013 [5] which used convolutional networks to extract features from region proposals instead of relying on hand-crafted low-level features such as edges, gradients and corners to detect objects. Several improvements were made over the next two years with the introduction of Faster R-CNN [17] which focused on improving the accuracy and speed of R-CNN. More recently, approaches have moved from two-stage architectures (CNN + RPN) to single-stage architectures (CNN) such as RetinaNet [10] and YOLOv3 [16] to improve the speed of the network while maintaining similar architectures. Due to the advent of these accurate and fast object detectors, tracking objects through detection has become a well instated method for tracking and counting fruit. Most approaches are generally formulated by combining object detectors, motion models, appearance models and data association algorithms.

Separation of target objects from distractors such as false positives is a challenging part of the association problem in the detect-to-track paradigm. At the time of publication, previous work focusing on simple tracking architectures have achieved state-of-the-art results. IoU Tracker [3] uses simple intersection over union matching as the cost for matching objects across frames, however it requires an extremely high frame rate detector. SORT [2] used the Hungarian data association algorithm with a Kalman filter cost to model the motion of defections and was at the time the best in class online tracking method. DeepSORT [19] improved on SORT by adding a cosine similarity cost between appearance vectors from a re-identification deep network used in the data association metric, while still running in real time.

This paradigm is applied to Mangos in [11] where a Faster R-CNN model detects regions of interest and a combination of Kanade-Lucas-Tomasi (KLT) optical flow estimator, Kalman Filters, and the Hungarian Assignment algorithm are used to associate detections to tracks, finally a structure from motion (SfM) algorithm is applied to reduce the effect of double-counting achieving an overall mean error of 27.8. Oranges and apples are counted per frame in [4] utilising

a neural network to produce detections and to regress the final image count, resulting in an overall $L2$ error (least square errors) of 13.7 and 10.5 for oranges and apples. The authors in [12] extend this work from counting fruit in single images to sequences of images. The proposed pipeline integrates deep neural network detections with SfM algorithms to count fruit from a single camera, achieving a $L1$ error (least absolute deviations) of 203 for oranges and 322 for apples.

3 Methods

This section describes our solution to counting rows of fruit in table-top farm environments autonomously from a mobile robotic platform with a mounted colour camera. We’re interested in obtaining the total fruit count (flower, un-ripe, ripe) per row for the purpose of informing farmers decisions on yield and labour. To count fruit, our solution aims to associate bounding boxes between cameras frames where the total number of unique associated detection IDs provides the total count. The basis of our approach is inspired by DeepSort [19] a tracking framework from the Multi-Object Tracking (MOT) challenge [14], that has outperformed much more complex architectures on the MOT benchmark. We introduce a set of novel components to bolster the tracking accuracy, (1) we update the CNN architecture (2) we add a novel classification branch to deal with multi-class data and force more distinct embeddings in the appearance feature space (3) we integrate robot odometry data (position along row) into the Kalman filter state space and (4) we augment the input data with contextual and aspect preserving qualities. The proposed pipeline is shown in Fig. 2. We detail the various changes in the following section, at its core the problem formulation as in [12] is for a set of images $\mathbf{I} = (I_k)_{k=1}^n$ containing n consecutive frames collected from a moving robotic platform and $c \in N$ fruit we are looking to find the mapping between the true fruit count c and estimated count $\hat{c}$.

3.1 Tracking

Our formulation for track handling and Kalman filtering is similar to SORT [2] and other MOT benchmarks. We assume no ego-motion information is available, but odometry information relating to robot position along a fruit row is provided. Note that the camera is not calibrated. We only utilise standard Kalman filters with a constant velocity motion model. The Kalman filter state space is defined as the 10 dimensional vector $(u, v, \gamma, h, r, \dot{u}, \dot{v}, \dot{\gamma}, \dot{h}, \dot{r})$ where (u, v) are the bounding box co-ordinates, (γ, h) are the aspect ratio and height and finally (r) is the row position of the robot. Row position r is a unit length from the start of robot operation at the start of the row. Co-ordinates (u, v, γ, h, r) are the respective object state and the following are respective velocities in image coordinates.

To accurately solve assignment between newly arrived detections and existing tracks, we formulate our approach into an assignment problem solvable

by the Hungarian algorithm. We use multiple models to represent motion, appearance and class description. Motion is incorporated in the model as squared Mahalanobis distance between predicted Kalman states dictated by $d^{(1)}$ in Eq. 1, where the i -th track distribution projection is denoted in measurement space as $(\mathbf{y}_i, \mathbf{S}_i)$ and the j -th new observation $\mathbf{d}_j$. Using this metric with a 10-dimensional measurement state space allows us to easily filter highly improbably associations with the 95% quartile of the Chi-square distribution $t^{(1)} = 11.07$ with 10 degrees of freedom, metrics are admissible if they are within this threshold by the indicator from Eq. 2 where $x = 1$.

$$d^{(1)}(i, j) = (\mathbf{d}_j - \mathbf{y}_i)^T \mathbf{S}_i^{-1} (\mathbf{d}_j - \mathbf{y}_i) \quad (1)$$

$$z_{i,j}^{(x)} = 1[d^{(x)}(i, j) \leq t^{(x)}] \quad (2)$$

Mahalanobis distance for multi-object data association might fail in situations while tracking on a mobile robot, such as when small angular/Cartesian movements of the camera are introduced. This can result in large shift in position in image space. Due to this, we incorporate two more models for improving the assignment problem. From the baseline implementation in [19] we keep the appearance descriptor shown in Eq. 3 and extended in Eq. 4 where the smallest cosine distance between i -th tracks and j -th bounding boxes is measured. For the appearance descriptor, r_j we keep a gallery $\mathbf{R}_k = \{\mathbf{r}_k^{(i)}\}_k^{O_k}$ of the last $O_k = 100$ observations for each track k . We extend the model by adding a new class description metric that considers the probability of different maturity stages. For a given observation with constant motion, occlusion can alter the intra-class appearance (i.e. covering the red part of a Strawberry may make it appear as unripe instead of ripe), to ensure consistent re-identification we compute a class description of all previous class observations.

$$d^{(2)}(i, j) = \min\{1 - \mathbf{r}_j^T \mathbf{r}_k^i \mid \mathbf{r}_k^i \in \mathbf{R}_i\} \quad (3)$$

We compute the class description metric w_j from the map of label probabilities of the classification head projected onto the unit hyper-sphere of previous class observations where $\|\mathbf{w}_j\| = 1$, similarly to computing the appearance metric, we utilise a gallery $\mathbf{W}_k = \{\mathbf{w}_k^{(i)}\}_k^{O_k}$ of previous observations of each track k , trivially inconsistent tracks before weighting with similar appearance but differing classes the total metric will be $\mathbf{w}_j + \mathbf{r}_j \leq \mathbf{r}_j + 1$. Eq. 4 denotes the cosine distance between label probabilities and previous label probability observations.

$$d^{(3)}(i, j) = \min\{1 - \mathbf{w}_j^T \mathbf{w}_k^i \mid \mathbf{w}_k^i \in \mathbf{W}_i\} \quad (4)$$

Combining the metrics ensures the association is more robust, taking into account motion, appearance and class description. As with squared Mahalanobis distance a binary variable is introduced to discount improbable associations for both appearance $t^{(2)}$ and class description $t^{(3)}$ in Eq. 2. The association problem cost matrix m is the combination of all the metrics with a weight parameter λ

where $x = 3$ and the association is admissible if it is within the gating region in Eq. 6.

$$m_{i,j} = \lambda d^{(1)}(x-1) + \sum_{p=2}^x (1-\lambda) d^{(p)}(i,j) \quad (5)$$

$$z_{i,j} = \prod_{p=1}^3 b^{(p)} \quad (6)$$

Suitable thresholds $t^{(x)}$ for each gating function $z_{i,j}^{(x)}$ can be found. For the motion metric chi-square distribution threshold will remove unlikely associations, for the appearance metric values of cosine similarity of similar bounding boxes from training data can be computed to find a suitable threshold and finally for the class description metric values $t^{(3)} \leq 1$ is a suitable value where closer to zero forces more consistent label probability history. The weight variable λ can be optimised depending on the domain, small values will prioritise appearance and class description over motion and higher values close to 1 will use mostly motion information to calculate the association and gating cost. We use the same matching cascade as in [19] extended with the new cost matrix m defined in Eq. 5.

3.2 Re-Identification and Class Description Network

To discriminate between different identities, we train a recurrent neural network to generate feature vectors $\mathbf{r}_j$ and $\mathbf{w}_j$ that minimise the cosine similarity between j -th bounding boxes of the same instance. The feature extraction stem architecture is described in Fig. 3. To train the network, we add two classification heads. The re-identification classification head attempts to map the identities vector $\mathbf{r}_j$ to the ground truth instance ID. The objective loss is Cosine Soft Max (Cross Entropy Loss). The label classification head maps $\mathbf{r}_j$ to label probabilities $\mathbf{w}_j$, Cross Entropy Loss is used to minimise the most probable class to the ground truth maturity stage (flower, unripe, ripe). The training procedure attempts to minimise both losses. For a batch size of 128 one forward pass of the network takes $412\mu\text{s}$ per bounding box on a modern GPU (Nvidia GeForce GTX 3090) making the method suitable for online tracking, with a max capability of > 2400 bounding boxes per second.

To deal with multi-class tracking as well as multi-object, we added the label classification head in order to coerce more separable features depending on fruit maturity stage. Fig. 4 visualises the Principal Component Analysis of learned feature vectors from a trained network with (bottom) and with (top) this extra label classification step. We visualise 128 dimension feature vectors as two principal components and colour the x, y components with the label class. It can be seen adding the extra network head creates much more separable features. To visualise the component space we plot a grid of 100 points around the extrema regions of principal components then take the PCA inverse transform

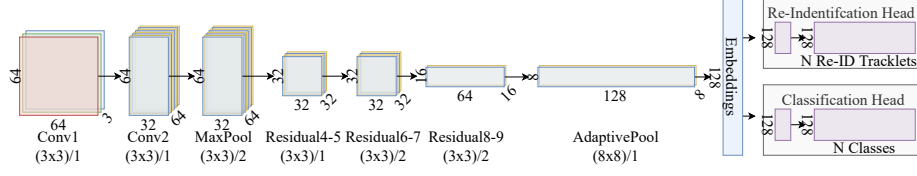


Fig. 3. Recurrent Neural Network Re-ID feature extractor. Input is a 64x64 patch of a bounding box detection and output is a 128 length feature vector projected onto the unit hyper sphere (for use with cosine similarity metric $d^{(2)}$). Each convolutional block (1, 2) is a 2D convolution followed by 2D batch normalisation and ReLU activation function. Each residual block (4, 5, 6, 7, 8, 9) is a basic ResNet block [7]. There are two heads to coerce more separable feature embeddings while training, a re-identification head and maturity classifier head. Total trainable parameters are 825,152.

of u , v components to get a representative feature vector. Once we have the representative features, we build a k-d tree of the original feature vectors and take the smallest cosine distance to the representative feature vectors to visualise an example bounding box (shown on the right) from the low dimensional u , v components. Much tighter and logical groupings are observed and the weight of different labels is more uniform, creating a more stable cosine distance between classes.

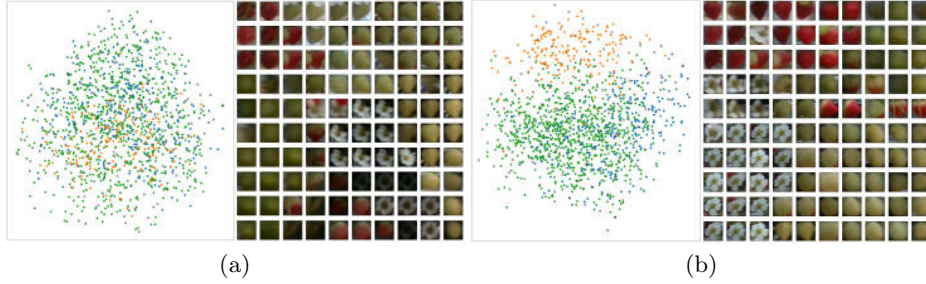


Fig. 4. Principal Component Analysis (left) and inverse mapping of 10 by 10 grid points (right, denoted by black crosses). Best viewed online. Analysis performed on the re-identification network feature vector of (a) baseline network and (b) improved network with classification head. The PCA (left) shows us the classification network (b) has better separation between classes (green=flower, blue=unripe, red=ripe) when the inverse of the PCA function is applied to the X and Y positions of each grid point.

3.3 Tracking Sequences

Image sequences of commercial strawberry plants were collected at the University of Lincoln's research farm at Riseholme, UK from a RGBD camera (Intel

RealSense D435i) mounted on the agricultural robot Thorvald [6] (see Fig. 1). The robot was deployed in two 8x24m poly-tunnels containing 5 table-top rows separated by a distance of 1.5m. The central row of each tunnel was used to capture the *Driscoll's Amesti* and *Driscoll's Katrina* data sequences. Sequences were collected one day apart late in the season (September) with the camera's height aligned to the strawberry soil bags. The data was acquired at 7Hz with the robot traversing the rows at 0.1ms (*Amesti-1*, *Katrina-1*, *Katrina-2*) and 0.2ms (*Amesti-2*) at a resolution of 1920×1080 . Data was annotated manually with expert knowledge into 3 distinct classes: Ripe ($> 85\%$ red coverage), Flower (white petals with no calyx shown) and Unripe (small to large immature green calyx visible). Each of the sequences capture was stopped at 500 frames. These images were captured and aligned using the [RaymondKirk/topic_store](#) and [IntelRealSense/realsense-ros](#) packages hosted on GitHub.

Four sequences were collected to validate our approach, containing two sides of table-top poly tunnel grown Strawberries of Amesti and Katrina variety. We will refer to the sequences in the format of *Variety-Side* where 1 and 2 are left and right sides of each row respectively. Amesti1-2 and Katrina 1-2 contain 12219, 4895, 15850 and 14507 bounding box annotations with 233, 172, 299 and 326 tracklets (count) respectively. To evaluate our proposed method, we define a *training* and *testing* split of a 75% to 25% ratio, to ensure no bias in evaluation of the system and promote generalisation when tracking. The training split consists of the images from *Katrina-1*, *Katrina-2* and *Amesti-2* sequences, whereas the testing split consists of *Amesti-1* image sequences. The training set was then split again by 75% and 25% to serve as training and validation data for optimising the re-identification model. The splits were chosen this way to ensure no bias in the final testing set used for evaluation. Some experiments contain data augmentation, we apply two different types which we deem square and padding, these transformations given in Eq. 7 and 8 of the raw bounding boxes are to preserve aspect ratio of textual features and embed surrounding environment context respectively. We share the data to support bench-marking in the community, since we found no other sequences openly available.

$$s(x, y, w, h) = \{ (x, (y + \frac{h}{2}) - \frac{w}{2}, w, w) | w > h, ((x + \frac{w}{2}) - \frac{h}{2}, y, h, h) | w \leq h \} \quad (7)$$

$$pad(x, y, w, h, p) = (x - p, y - p, w + 2p, h + 2p) \quad (8)$$

3.4 Evaluation Metrics

Let $f : \mathbf{I}^{(j)} \rightarrow c \in N$ for a sequence $\{\mathbf{I}^{(j)}\}_{j=1}^{500}$ we evaluate our proposed methods as the Least Absolute Deviation ($L1$ loss) of $c^{(j)}$ and $\hat{c}^{(j)}$.

$$\sum_{j=1}^{500} |f(\mathbf{I}^{(j)}) - \hat{c}^{(j)}| \quad (9)$$

For specific classes, we consider only c , $\hat{c}$ fruit belonging to the specific class and ignore others. We use this loss in model selection and to validate the proposed system. Specifically Eq. 9 concerns the testing set *Amesti-1* for evaluation.

4 Results

In this section, we now evaluate our counting-by-tracking pipeline. We first detail the training regime for each experiment. We then compare the baseline system to seven other experiments with the following modifications: (1) addition of a label probability cosine cost $d^{(3)}$ and label classification head to the re-identification network (2) detection augmentations *square* in Eq. 7 and *pad* in Eq. 8 and (3) a combined improvements network consisting of the all modifications that resulted in an improved score shown in Fig. 5. We apply all evaluations in this section to the hand labelled *Amesti-1* data sequence which was hidden during network training against the predicted count data per-frame and total per-sequence.

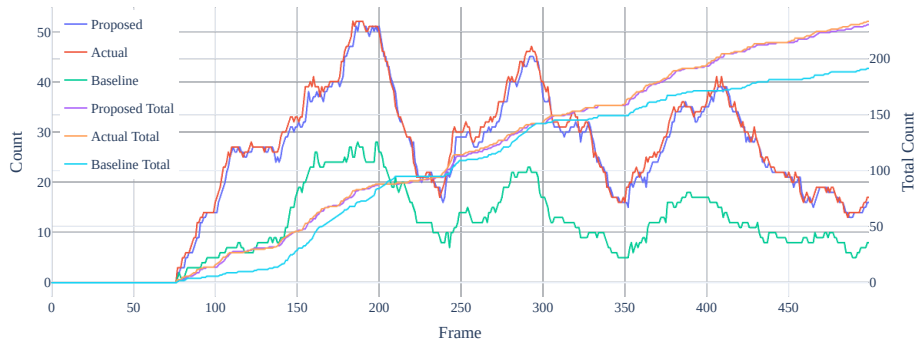


Fig. 5. Counting-by-tracking performance of the baseline approach and our proposed combined method, detailed in Table 1. Total counts over time are given, and also frame by frame counting results for all classes combined (class agnostic tracking). It can be seen our proposed system (dark blue, purple) performs much better and achieves almost perfect frame-by-frame counting. Whereas the baseline system (light blue, green) has a much lower overall count accuracy.

To train the models we input a batch of 128 bounding boxes per iteration and train for a total of 6400 iterations, reducing the base LR of 0.1 by a factor of 10 at 80% and 90% of total iterations. To ensure consistent results, we apply the data augmentation to the data directly before training and output the re-identification accuracy and label classification accuracy (relevant only for the Sub-Net experiment) every 400 iterations. The re-identification accuracy during training for the baseline model and Sub-Net model were 97% and for all other experiments the accuracy was >99% when rounded to two significant figures.

Table 1 describes the results of the experiments, we present the results per-class and class agnostic since intrinsically the baseline tracking system is class

agnostic. The results show a strong improvement over the baseline experiment across all experiments. Bold values of $L1$ loss indicate the best experiment for each evaluation metric.

Table 1. $L1$ Loss of Berry Count on Amesti-1

Experiment	Class			
	Agnostic	Flower	Ripe	Unripe
Baseline	41 (192/233)	1 (23/24)	19 (69/50)	59 (100/159)
Maintaining Aspect Ratio				
Square	57 (176/233)	2 (26/24)	3 (53/50)	62 (97/159)
Classification Network Cost Matrix				
Sub-Net	31 (202/233)	1 (25/24)	3 (53/50)	35 (124/159)
Embedding Context in Detection				
Pad-8	33 (266/233)	14 (38/24)	29 (79/50)	10 (149/159)
Pad-16	86 (319/233)	13 (37/24)	36 (86/50)	37 (196/159)
Pad-32	25 (258/233)	3 (27/24)	12 (62/50)	10 (169/159)
Pad-64	8 (241/233)	3 (27/24)	4 (54/50)	1 (160/159)
Combined Improvements				
Combined	7 (240/233)	2 (26/24)	2 (52/50)	3 (162/159)

5 Conclusion

We have presented a framework for accurately tracking and counting fruit in a complex scene from bounding boxes, extending on current tracking architecture. Our system utilises re-identification features and label probability vectors with cosine similarity as well as robot odometry formatted as row position and data augmentation methods to maintain track consistency through occlusions and to maintain tracks in densely clustered detection regions.

The results demonstrate that our system is capable of reliably tracking and counting multiple classes in clusters from multiple view points. We use an off the shelf cheap computer vision camera and modern GPUs so that our system can be applied easily. Our results indicate an improvement of $L1$ loss from 41, 1, 19 and 59 (all classes, flower, ripe, unripe) to 7, 2, 2, 3 error in counting 500 frames of the *Amesti-1* sequence not used in training. On a modern GPU we are able to process >2400 bounding box detections per-second in a single forward pass of the network, enabling online tracking applications.

Future work will include further evaluation of the proposed system and further ablation studies into tracking by using additional cues, utilising different feature extraction networks, class embeddings and extending to full 3D space with approaches such as depth data integration or recent structure from motion approaches. We applied our counting system to soft-fruit, however the generic nature of the proposed solution makes it applicable to a wide range of object

counting applications beyond the soft-fruit scenario where a mobile robot can operate.

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